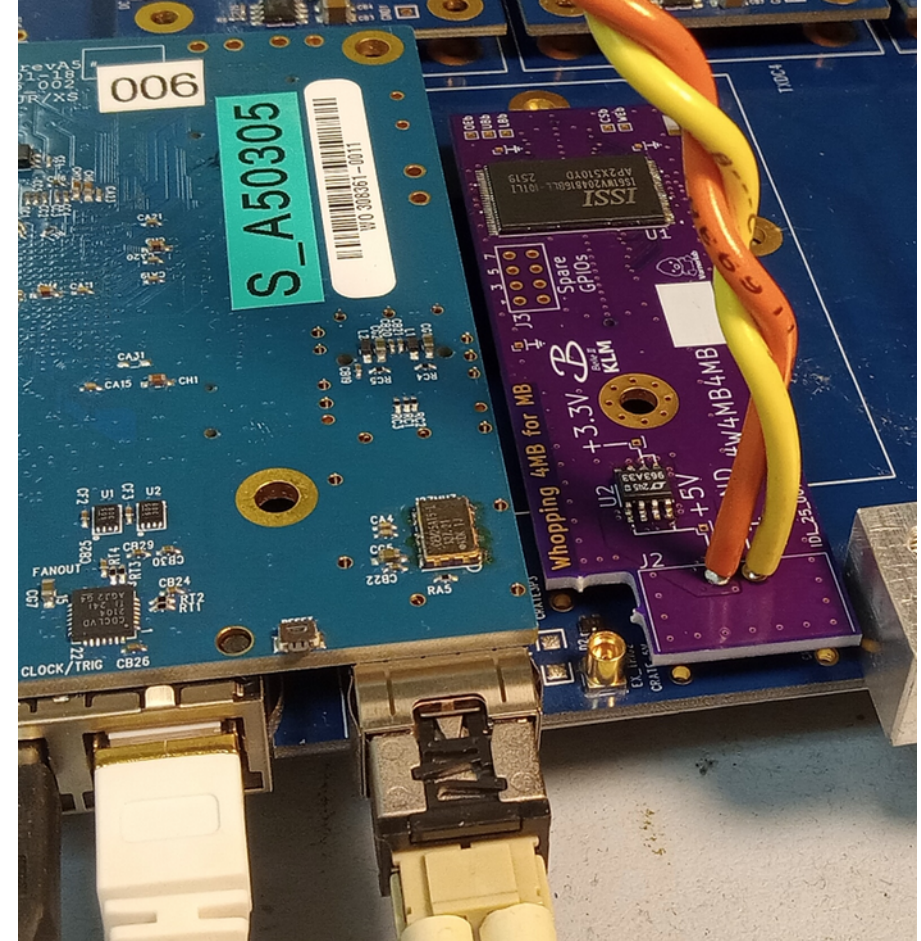
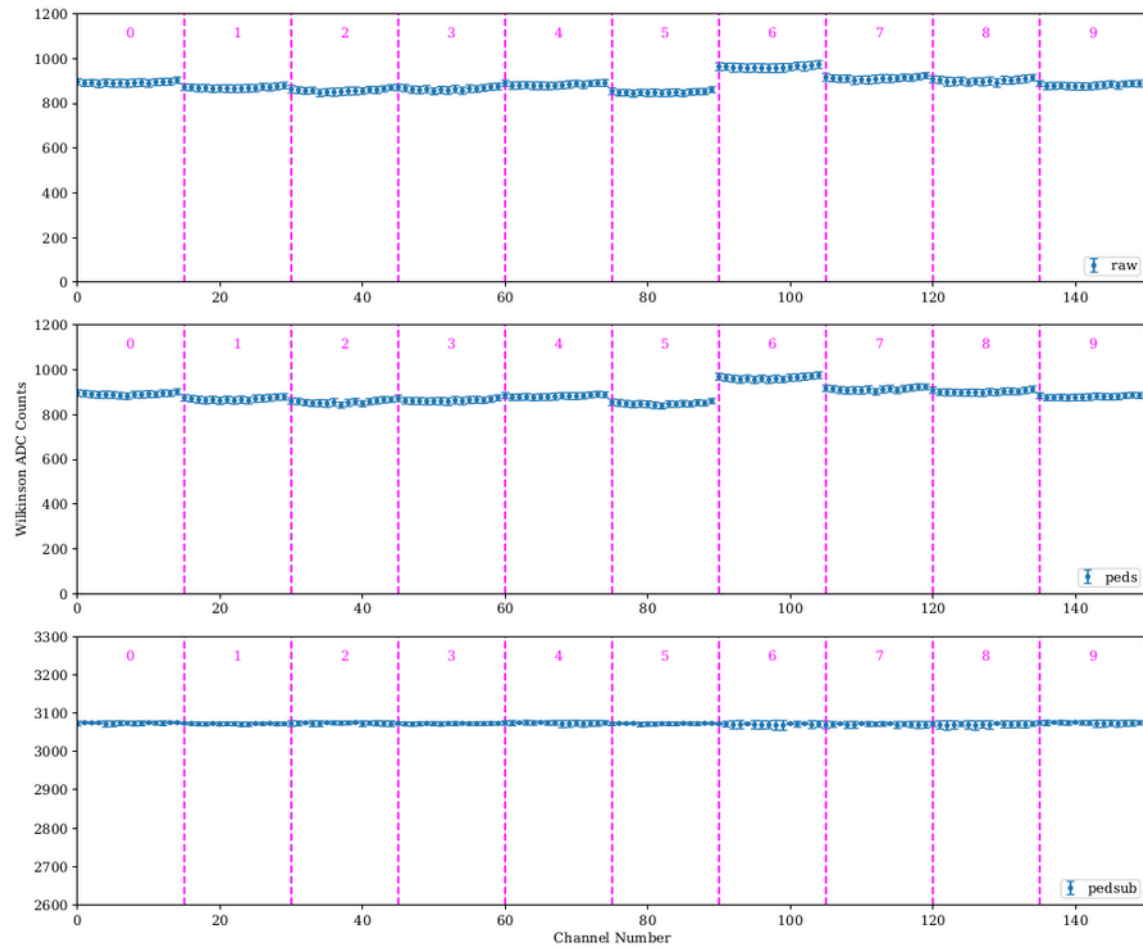


SRAM UPDATE

- ☕ Last time: Programmed lane 1 SCROD with bitfile klm_scint_v3201.bit compiled last July --- New SRAM not working
- ☕ Probed testpoints
 - ☕ Nothing measured as expected
- ☕ Found problem
 - ☕ klm_scint_v3201.bit was written before the board was made --- we did not know the pinout at that time
 - ☕ Now works great



TEST RESULTS



VERSION LOG



v3201



First compile of SRAM upgrade firmware



v3202



Tied **BUS_REGCLR** to GND as in master branch



v3203



Fix pin mapping in UCF file



v3301



Merge with vFE33 and
feature/ramp_settle

- **ramp_settle** feature not working as anticipated

VERSION LOG



v3201



First compile of SRAM upgrade firmware



v3202



Tied **BUS_REGCLR** to GND as in master branch



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Fix pin mapping in UCF file



v3301



Merge with vFE33 and
feature/ramp_settle

- **ramp_settle** feature not working as anticipated



Comments on **feature/ramp_settle**



Symptom was lower than nominal ADC counts for 2nd digitization window whenever 1st digitization window only shifted out last 4 samples



Intention was to delay **BUS_RAMP** signal if it did not have enough settle time from the last activation



Result is higher than nominal ADC counts for 2nd digitization window under same conditions as above



Hypothesis: so-called **BUS_RD_ENA** signal actually starts gray-code counter



I will either move the ramp delay up the FW hierarchy or encapsulate the **BUS_RD_ENA** signal under the same delay logic --- whichever is cleaner

REMAINING TASKS

SRAM Expansion PCB

-  Finalize +5V connector choice
(Cheap, easy, secure, non-interference)
-  Double-check PCB design
(Characteristic impedance? Image-current path? Happy with silkscreen?)
-  Order 3 more or order full production run

Design RTM board for +5V supply

Design wire harness (Determine gauge from power budget, appropriate flexibility or stiffness, how to sheath and preform bends)

Investigate feasibility of securing SRAM PCB with standoff

-  Enough threads for proper engagement?
-  Necessary screw sizes available to fully
utilize threads?
-  Screw heads don't interfere with neighbor
boards?

Investigate SCROD fastening hardware at same time

-  Same considerations as above
-  Ease of installation?

Order all hardware, wire spools, and ICs

Design clamp tool for easy MB removal

OTHER ACTIVITIES



Booked all my KLM shifts for 2026a



all in next few weeks



Make slides for KLM group meeting next week



Travel to Japan? (Should I?)



Way behind on CR shifts



May want to test-fit prototype RTM power supply board and wire harness before constructing all units